NZ SPORTS TURF INSTITUTE
Since 1949

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R J Hill Laboratories Limited
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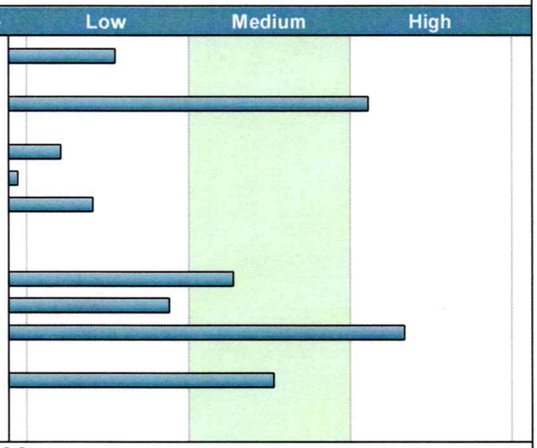
Certificate of Analysis

Client:	Russley Golf Club	**Lab No:**	2606324 shvpv2
Address:	PO Box 14045 Christchurch Airport Christchurch 8053	**Date Received:**	07-May-2021
		Date Reported:	13-May-2021 **(Amended)**
		Quote No:	
		Order No:	
		Client Reference:	Mike Yorston
Phone:	03 358 5051	**Submitted By:**	Chris Gribben

Sample Name: Green 1 **Lab Number:** 2606324.1
Sample Type: TURF Browntop, Sand (S279)

Analysis		Level Found	Medium Range
pH	pH Units	4.7	5.0 - 5.7
Olsen Phosphorus	mg/L	18	5 - 15
Potassium	me/100g	0.12	0.20 - 0.50
Calcium	me/100g	0.5	2.0 - 4.0
Magnesium	me/100g	0.18	0.30 - 0.70
Sodium	me/100g	< 0.05	0.00 - 0.20
CEC	me/100g	4	3 - 6
Total Base Saturation	%	23	25 - 60
Volume Weight	g/mL	1.30	0.60 - 1.20
K/Mg Ratio*		0.7	0.3 - 1.0
Soil Sample Depth*†	mm	0-75	



Base Saturation %	K 3.2	Ca 14	Mg 4.7	Na 0.9
MAF Units	K 3	Ca < 1	Mg 5	Na 2

The above nutrient graph compares the levels found with reference interpretation levels. NOTE: It is important that the correct sample type be assigned, and that the recommended sampling procedure has been followed. R J Hill Laboratories Limited does not accept any responsibility for the resulting use of this information. IANZ Accreditation does not apply to comments and interpretations, i.e. the 'Range Levels' and subsequent graphs.

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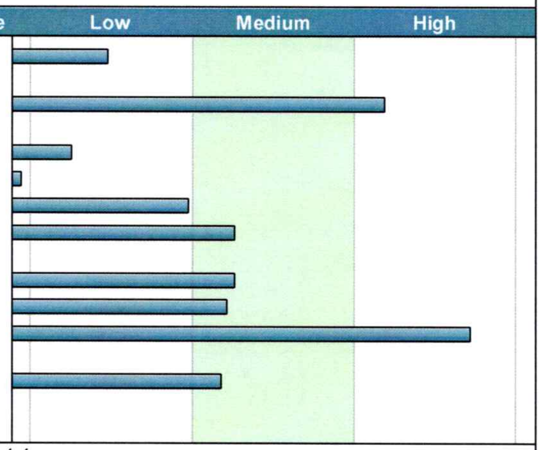
Phone:	03 358 5051	**Submitted By:**	Chris Gribben

Sample Name: Green 13 **Lab Number:** 2606324.2
Sample Type: TURF Browntop, Sand (S279)

Analysis		Level Found	Medium Range
pH	pH Units	4.7	5.0 - 5.7
Olsen Phosphorus	mg/L	20	5 - 15
Potassium	me/100g	0.12	0.20 - 0.50
Calcium	me/100g	0.8	2.0 - 4.0
Magnesium	me/100g	0.30	0.30 - 0.70
Sodium	me/100g	0.05	0.00 - 0.20
CEC	me/100g	4	3 - 6
Total Base Saturation	%	32	25 - 60
Volume Weight	g/mL	1.42	0.60 - 1.20
K/Mg Ratio*		0.4	0.3 - 1.0
Soil Sample Depth*†	mm	0-75	
Base Saturation %		K 3.3 Ca 20 Mg 7.8 Na 1.4	
MAF Units		K 4 Ca 1 Mg 9 Na 3	



The above nutrient graph compares the levels found with reference interpretation levels. NOTE: It is important that the correct sample type be assigned, and that the recommended sampling procedure has been followed. R J Hill Laboratories Limited does not accept any responsibility for the resulting use of this information.
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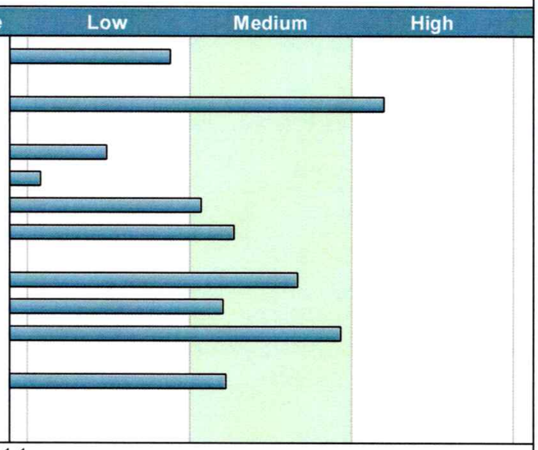
		Client Reference:	Mike Yorston	
Phone:	03 358 5051	**Submitted By:**	Chris Gribben	

Sample Name: Green 18 **Lab Number:** 2606324.3
Sample Type: TURF Browntop, Sand (S279)

Analysis		Level Found	Medium Range
pH	pH Units	4.9	5.0 - 5.7
Olsen Phosphorus	mg/L	20	5 - 15
Potassium	me/100g	0.15	0.20 - 0.50
Calcium	me/100g	1.1	2.0 - 4.0
Magnesium	me/100g	0.33	0.30 - 0.70
Sodium	me/100g	0.05	0.00 - 0.20
CEC	me/100g	5	3 - 6
Total Base Saturation	%	32	25 - 60
Volume Weight	g/mL	1.16	0.60 - 1.20
K/Mg Ratio*		0.5	0.3 - 1.0
Soil Sample Depth*†	mm	0-75	



Base Saturation %	K 3.0	Ca 21	Mg 6.6	Na 1.1
MAF Units	K 4	Ca 2	Mg 9	Na 3

The above nutrient graph compares the levels found with reference interpretation levels. NOTE: It is important that the correct sample type be assigned, and that the recommended sampling procedure has been followed. R J Hill Laboratories Limited does not accept any responsibility for the resulting use of this information.
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Phone:	03 358 5051	**Submitted By:**	Chris Gribben

Soil Analysis Results

Sample Name:		Green 1	Green 13	Green 18			
Lab Number:		2606324.1	2606324.2	2606324.3			
Sample Type:		TURF Browntop, Sand	TURF Browntop, Sand	TURF Browntop, Sand			
Sample Type Code:		S279	S279	S279			
pH	pH Units	4.7	4.7	4.9	-	-	-
Olsen Phosphorus	mg/L	18	20	20	-	-	-
Potassium	me/100g	0.12	0.12	0.15	-	-	-
Potassium	%BS	3.2	3.3	3.0	-	-	-
Potassium	MAF units	3	4	4	-	-	-
Calcium	me/100g	0.5	0.8	1.1	-	-	-
Calcium	%BS	14	20	21	-	-	-
Calcium	MAF units	< 1	1	2	-	-	-
Magnesium	me/100g	0.18	0.30	0.33	-	-	-
Magnesium	%BS	4.7	7.8	6.6	-	-	-
Magnesium	MAF units	5	9	9	-	-	-
Sodium	me/100g	< 0.05	0.05	0.05	-	-	-
Sodium	%BS	0.9	1.4	1.1	-	-	-
Sodium	MAF units	2	3	3	-	-	-
CEC	me/100g	4	4	5	-	-	-
Total Base Saturation	%	23	32	32	-	-	-
Volume Weight	g/mL	1.30	1.42	1.16	-	-	-
K/Mg Ratio*		0.7	0.4	0.5	-	-	-
Soil Sample Depth*†	mm	0-75	0-75	0-75	-	-	-

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Analyst's Comments

† Customer supplied data. Please note: Hill Laboratories cannot be held responsible for the validity of this customer supplied data, or any subsequent calculations that rely on this information.

Samples 1-3 Comment:
The medium or optimum range guidelines shown in the histogram report relate to sampling protocols as per Hill Laboratories' crop guides and are based on reference values where these are published. Results for samples collected to different depths than those described in the crop guide should be interpreted with caution.
For pastoral soils, the medium ranges are specific for a 75mm sample depth, but if a 150mm sampling depth is used the nutrient levels measured may appear low against these ranges, as nutrients are typically more concentrated in the top of the soil profile. These soil profile differences are altered upon cultivation or contouring.

Samples 1-3 Comment:
Cation results for turf soil samples are expressed as % base saturation levels of the C.E.C. Alternative units (either me/100g and MAF units) are given at the bottom of the graph.

Samples 1-3 Comment:
The low CEC level found in this soil indicates that it can only retain low levels of cation nutrients (potassium, calcium, magnesium and sodium). The bar graph format shown above presents the cation levels in terms of a % base saturation level, which allows for the soil's low cation retention.

Amended Report: This certificate of analysis replaces report '2606324-shvpv1' issued on 12-May-2021 at 3:31 pm.
Reason for amendment: Following an internal query, the pH analysis has been repeated and as a consequence, an error in the original result reported has been identified. The pH result for sample 2 (Green 13) has been amended from 5.3 to 4.7 and the pH result for sample 3 (Green 18) has been amended from 5.4 to 4.9.

Summary of Methods

The following table(s) gives a brief description of the methods used to conduct the analyses for this job. The detection limits given below are those attainable in a relatively simple matrix. Detection limits may be higher for individual samples should insufficient sample be available, or if the matrix requires that dilutions be performed during analysis. A detection limit range indicates the lowest and highest detection limits in the associated suite of analytes. A full listing of compounds and detection limits are available from the laboratory upon request.
Unless otherwise indicated, analyses were performed at Hill Laboratories, 28 Duke Street, Frankton, Hamilton 3204.

Sample Type: Soil

Test	Method Description	Default Detection Limit	Sample No
Sample Registration*	Samples were registered according to instructions received.	-	1-3
Soil Prep (Dry & Grind)*	Air dried at 35 - 40°C overnight (residual moisture typically 4%) and crushed to pass through a 2mm screen.	-	1-3
pH	1:2 (v/v) soil:water slurry followed by potentiometric determination of pH. In-house.	0.1 pH Units	1-3
Olsen Phosphorus	Olsen extraction followed by Molybdenum Blue colorimetry. In-house method.	1 mg/L	1-3
Potassium	1M Neutral ammonium acetate extraction followed by ICP-OES. In-house.	1 MAF units	1-3
Calcium	1M Neutral ammonium acetate extraction followed by ICP-OES. In-house.	1 MAF units	1-3
Magnesium	1M Neutral ammonium acetate extraction followed by ICP-OES. In-house.	1 MAF units	1-3
Sodium	1M Neutral ammonium acetate extraction followed by ICP-OES. In-house.	2 MAF units	1-3
Potassium	1M Neutral ammonium acetate extraction followed by ICP-OES. In-house.	0.01 me/100g	1-3
Calcium	1M Neutral ammonium acetate extraction followed by ICP-OES. In-house.	0.5 me/100g	1-3

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Sample Type: Soil

Test	Method Description	Default Detection Limit	Sample No
Magnesium	1M Neutral ammonium acetate extraction followed by ICP-OES. In-house.	0.04 me/100g	1-3
Sodium	1M Neutral ammonium acetate extraction followed by ICP-OES. In-house.	0.05 me/100g	1-3
Potassium	1M Neutral ammonium acetate extraction followed by ICP-OES. In-house.	0.1 %BS	1-3
Calcium	1M Neutral ammonium acetate extraction followed by ICP-OES. In-house.	1 %BS	1-3
Magnesium	1M Neutral ammonium acetate extraction followed by ICP-OES. In-house.	0.2 %BS	1-3
Sodium	1M Neutral ammonium acetate extraction followed by ICP-OES. In-house.	0.1 %BS	1-3
CEC	Summation of extractable cations (K, Ca, Mg, Na) and extractable acidity. May be overestimated if soil contains high levels of soluble salts or carbonates. In-house.	2 me/100g	1-3
Total Base Saturation	Calculated from Extractable Cations and Cation Exchange Capacity.	5 %	1-3
Volume Weight	The weight/volume ratio of dried, ground soil. In-house.	0.01 g/mL	1-3

These samples were collected by yourselves (or your agent) and analysed as received at the laboratory.

Testing was completed between 08-May-2021 and 13-May-2021. For completion dates of individual analyses please contact the laboratory.

Samples are held at the laboratory after reporting for a length of time based on the stability of the samples and analytes being tested (considering any preservation used), and the storage space available. Once the storage period is completed, the samples are discarded unless otherwise agreed with the customer. Extended storage times may incur additional charges.

Karen Currie BSc
Client Services Manager - Agriculture